Object Oriented Programming

Java

Socket Programming

What Is a Socket?

- A socket is one **end-point** of a **two-way communication link** between two programs running on the network.
- Socket classes are used to represent the connection between a client program and a server program.
- The java.net package provides two classes :
  - **Socket:** implements the client side of the connection
  - **ServerSocket:** implements server side of the connection.

What Is a Socket?

- A socket is bound to a port number so that the TCP layer can identify the application that data is destined to be sent to.
- Normally, a server runs on a specific computer and has a socket that is bound to a specific port number.

<IP address,   Port number>

- The server just waits, listening to the socket for a client to make a connection request.
- **On the client-side:** The client knows the **hostname** of the machine on which the server is running and the **port number** on which the server is listening.

What Is a Socket?

- When the connection establishes, the client and server can now communicate by writing to or reading from their sockets.
- An endpoint is a combination of an IP address and a port number.
- Every TCP connection can be uniquely identified by its two endpoints.
- That way you can have multiple connections between your host and the server.

Sockets

- Socket types:
  - STREAM – uses TCP which is reliable, stream oriented protocol
  - DATAGRAM – uses UDP which is unreliable, message oriented protocol
  - RAW – provides RAW data transfer directly over IP protocol (no transport layer)
- Sockets can use
  - “unicast ” (for a particular IP address destination)
  - “multicast” (a set of destinations – 224.x.x.x)
  - “broadcast ” (direct and limited)
  - “Loopback ” address i.e. 127.x.x.x

Ports

- Each host has **65,536** ports
- Some ports are reserved for specific apps
  - 20, 21: FTP
  - 23: Telnet
  - 80: HTTP

- A socket provides an interface to send data to/from the network through a port

Ports

- Port numbers are used to identify services on a host
- Port numbers can be:
  - well-known
  - dynamic or private
  - Clients usually use dynamic ports

Basic steps of socket programming

Server Side

- Import necessary packages

```
import java.io.*;
import java.net.*;
```

- Create a server socket and bind it to a specific port

```
ServerSocket serverSocket = new ServerSocket(12345);
```

- Accept incoming client connections

```
Socket clientSocket = serverSocket.accept();
```

- Get input and output streams to send and receive data

```
InputStream input = clientSocket.getInputStream();
OutputStream output = clientSocket.getOutputStream();
```

Basic steps of socket programming

Client Side

- Import necessary packages

```
import java.io.*;
import java.net.*;
```

- Create a socket and connect it to the server's IP address and port

```
Socket socket = new Socket("server_ip_address", 12345);
```

- Get input and output streams for communication

```
InputStream input = socket.getInputStream();
OutputStream output = socket.getOutputStream();
```

Basic steps of socket programming

- Import necessary packages
- Create a server socket and bind it to a specific port
- Accept incoming client connections
- Get input and output streams to send and receive data